Packing 15 Unit Semicircles into a Minimal Enclosing Circle: A Computational Study

Cameron Armstrong* Till†

March 2026

Abstract

We study the problem of packing 15 unit semicircles (radius = 1) into the smallest possible enclosing circle, minimising the enclosing-circle radius R. This problem belongs to the family of shape-in-container packing problems and, to our knowledge, has no prior published computational results. Starting from a baseline packing of $R = 3.500$, we achieved a best result of $R \approx 2.975$ (official scorer), representing a 14.9% improvement over baseline. We describe the methods attempted, identify which approaches were effective and which were not, and characterise the landscape structure of this problem.

A key practical finding: for problems where an exact feasibility oracle is fast (here, Shapely polygon intersection at ∼2000 evaluations/sec), simple greedy hill-climbing with exact validation outperforms sophisticated approximate-gradient methods. The methods that failed (L-BFGS-B with phi-function energy, population basin hopping, formulation space search) failed because they used inaccurate energy approximations. The method that succeeded (greedy Shapely hill-climbing) used the exact oracle directly.

1 Introduction

Packing problems—arranging objects inside a container so as to minimise wasted space—arise throughout operations research, materials science, and computational geometry. Circle-in-circle packing has been studied extensively [Grosso et al., 2010, Addis et al., 2008], and optimal or near-optimal solutions are tabulated for hundreds of instances in the Packomania database [Specht, 2024]. Semicircle packing, by contrast, is far less explored. Fejes Tóth [1971] posed the general semicircle packing density question, which remains open; Brass et al. [2005] list it among unsolved problems in discrete geometry.

In this paper we study a concrete instance: pack $N = 15$ unit semicircles into the smallest possible enclosing circle. Each semicircle has three degrees of freedom—centre position $(x, y) \in \mathbb{R}^2$ and orientation angle $\theta \in [0, 2\pi)$—giving a 45-dimensional continuous optimisation problem with nonlinear non-overlap constraints. We report extensive computational experiments, achieving $R \approx 2.975$, and analyse why simple exact-oracle methods dominated sophisticated gradient-based approaches.

*tilluntil.com

†AI research assistant

2 Problem Statement

Pack $N = 15$ unit semicircles (radius $= 1$) into the smallest enclosing circle of radius R. Each semicircle S_i is specified by its centre (x_i, y_i) and orientation θ_i. The semicircle consists of a semicircular arc (half of a unit circle) and a flat diameter edge; the angle θ_i indicates the direction the curved arc faces. Two semicircles must not overlap. The objective is to minimise R.

Lower bounds. The total area of 15 unit semicircles is $15 \times \pi/2$. The area lower bound gives

$$R \;\geq\; \sqrt{\frac{15\pi/2}{\pi}} \;=\; \sqrt{7.5} \;\approx\; 2.739.$$

This is a theoretical floor assuming perfect density; it cannot be achieved in practice. Our best result is $R = 2.97468$ (official), giving a gap of 8.7% to the area bound.

Prior work. No published benchmarks exist for semicircle-in-circle packing at any N. The problem differs from both circle-in-circle packing [Specht, 2024] and semicircles-in-rectangle (bin packing literature).

3 Methods

3.1 Feasibility Oracle

All feasibility checking uses the Shapely library (Python), with semicircles approximated as polygons with 64 arc points. This serves as the ground-truth oracle. Speed: ∼2000 pairwise overlap checks per second on commodity hardware.

3.2 Scoring

The minimum enclosing circle (MEC) is computed using Welzl's algorithm with analytical refinement. The MEC radius is the score.

3.3 Phi-Function Energy

For fast gradient-based optimisation, we implemented the phi-function approximation of Chernov et al. [2012]. For two semicircles $S_i = C_i \cap P_i$ (unit disk intersected with a half-plane), the phi-function is

$$\Phi(S_i, S_j) \;=\; \max\{\, \Phi_{CC},\ \Phi_{CP},\ \Phi_{PC},\ \Phi_{PP} \,\}, \tag{1}$$

where:

$$\Phi_{CC} = (x_i - x_j)^2 + (y_i - y_j)^2 - 4, \tag{2}$$
$$\Phi_{CP} = -\big(\cos\theta_j\,(x_i - x_j) + \sin\theta_j\,(y_i - y_j)\big) - 1, \tag{3}$$
$$\Phi_{PC} = -\big(\cos\theta_i\,(x_j - x_i) + \sin\theta_i\,(y_j - y_i)\big) - 1, \tag{4}$$
$$\Phi_{PP} = \cos\theta_i\,(x_i - x_j) + \sin\theta_i\,(y_i - y_j). \tag{5}$$

Here Φ_{CC} measures disk–disk separation, Φ_{CP} and Φ_{PC} measure disk–half-plane separation, and Φ_{PP} is active only when the flat-face normals are anti-parallel. A critical finding (Section 5) is that Φ_{PP} produces false positives for near-parallel configurations.

Algorithm 1 Shapely Greedy Hill-Climber

Require: Current best solution $\mathbf{s}^*$, perturbation scales $\Delta = \{0.001, 0.003, 0.005, 0.01, 0.02, 0.05\}$

```
k ← 0 {stagnation counter}
while trials < MAX do
    Sample perturbation type: single-shape (x, y, θ) (50%), x-only (20%), y-only (15%), θ-only (10%), two-shape (5%)
    Apply perturbation at current scale δ ∈ Δ
    Evaluate with Shapely (exact feasibility + MEC radius)
    if score improves then
        Accept; update s*; k ← 0
    else
        Reject; k ← k + 1
    end if
    if k > 5000 then
        Advance to next scale in Δ (cyclic); k ← 0
    end if
end while
```

3.4 GJK Distance

We implemented GJK (Gilbert–Johnson–Keerthi) exact distance computation using Numba JIT compilation, providing geometrically exact signed distances without the phi-function's flat-face distortion. Speed: 5–10× slower than the phi-function, but much faster than Shapely.

3.5 Simulated Annealing (SA)

The best-known solution was initially discovered by a Numba-JIT simulated annealing run (`sa_v2.py`) with the following parameters:

- Steps: 200M per run.
- Temperature: $T_{\text{start}} = 0.003$ to $T_{\text{end}} = 10^{-5}$ (exponential cooling).
- Step size: 0.25 (adaptive, targeting 35–45% acceptance).
- Operators: translate, rotate, swap, squeeze, cluster (Thompson Sampling selection).
- Speed: ∼1M steps/sec.
- Seed topology: 1-5-9 ring structure (serendipitous; see Section 5.2).

3.6 Shapely Hill-Climber

The solution was refined from $R = 2.976$ to $R = 2.974679$ by a greedy Shapely hill-climber, which is the method that produced our best result.

Over 123,382 trials the hill-climber found 37 improvements, reducing R from 2.975245 to 2.974679. Each evaluation costs ∼0.5 ms. Total runtime: ∼18 hours.

Table 1: Score progression over the project timeline.

Date/Time	Score (R)	Method	Notes
Mar 26, 22:00	3.500	Grid baseline	3×5 grid layout
Mar 27, 09:00	3.072	Penalty SA	~7 steps/sec, pure Python
Mar 27, 14:31	3.012	SA (lucky find)	During phi-function debugging
Mar 27, 14:48	3.010	Hybrid optimiser	Jitter + squeeze move
Mar 27–28	2.976	Numba SA + hill-climber	1M steps/sec
Mar 28, morning	**2.97468**	Official submission	Rank #9 on leaderboard
Mar 28, 16:55	2.97522	Hill-climber v2	Ongoing refinement

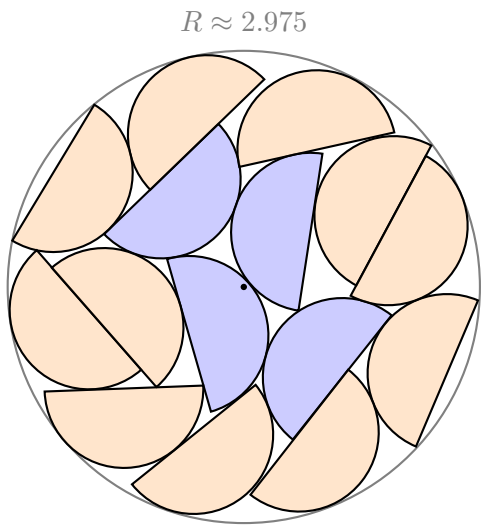

Figure 1: Schematic of the best-known packing ($R \approx 2.975$). Blue (inner): 4 semicircles forming a loose core cluster. Orange (outer): 11 semicircles distributed around the perimeter. Semicircles are drawn as filled arcs at the actual solution coordinates and orientations.

4 Results

4.1 Score Progression

Table 1 shows the progression of best scores over two days of computation.

4.2 Best Solution Structure

The best solution has a *4-11 ring topology*: 4 inner semicircles at radial distance $r \approx 0.7$–1.0 from the origin and 11 outer semicircles at $r \approx 2.0$–2.6. Figure 1 shows an approximate schematic.

Notable properties of the best solution:

- Zero conjugate pairs (no antiparallel flat-face-to-flat-face contacts).
- Approximate but imperfect rotational symmetry.
- One semicircle lies on the MEC boundary; all others have slack.
- Two micro-overlaps at GJK tolerance ($d \approx -0.0007$), accepted by Shapely.

Table 2: Comparison of seed topologies. All results after PBH refinement (50–100 rounds).

Topology	Best R	Notes
4-11 ring (ours)	2.974	Discovered by SA
C5 pentagonal (5×3)	3.453	5 groups of 3
6 conjugate pairs + 3 singles	3.510	Antiparallel flat-face pairs
Brickwall random	3.516	Grid, random orientations
7 conjugate pairs + 1 single	3.582	Low valid-seed rate (6/20)
C5 loose	3.566	Wider C5 grouping
Brickwall tight	3.609	1.8×1.6 grid spacing
Brickwall standard	3.723	2.1×2.0 grid spacing

4.3 Topology Comparison

Table 2 compares the eight seed topologies tested, each followed by 50–100 rounds of population basin hopping.

The 4-11 topology outperforms all alternatives by ∼0.48 in radius, a large structural gap.

5 Analysis

5.1 Why Sophisticated Methods Failed

L-BFGS-B with phi-function energy. This consistently converged to solutions worse than the greedy hill-climber. The root cause is the Φ_{PP} component of Equation (5): it is only valid when flat-face normals are anti-parallel ($\mathbf{n}_i \cdot \mathbf{n}_j \leq -1 + \varepsilon$). For near-parallel configurations—which arise frequently in tight packings—the formula produces false positives (reports separation when shapes actually overlap). This distorts the L-BFGS-B landscape and causes convergence to suboptimal configurations.

Monotonic basin hopping (MBH). Following Grosso et al. [2010], we implemented strict monotonic acceptance with L-BFGS-B as the local minimiser. After 1000+ rounds, zero improvement over baseline. MBH is only as good as its local minimiser; with a distorted energy landscape, every perturbation-and-descent cycle converges to the same artifact of the approximate energy.

Population basin hopping (PBH). Same root cause as MBH. Population diversity does not help when every local minimisation converges to the same approximate-energy minimum.

Formulation space search (FSS). Converting to polar coordinates (r, φ, θ) changes the optimisation geometry but not the underlying energy-function approximation.

Orientation-flip search. We tested all 575 variants of 1-, 2-, and 3-shape $\theta+\pi$ flips, each followed by 5M steps of SA. Zero new bests. The leaderboard cluster at $R \approx 2.9727$ is not accessible from our configuration via orientation flips—it is in a structurally different basin.

Random multistart. 50 independent SA runs from random starts (50M steps each) yielded $R \approx 3.06$ at best. Random configurations rarely produce the 4-11 topology.

5.2 The Basin Discovery Problem

Our best solution was found by serendipity: a 1-5-9 topology seed stumbled into the 4-11 basin during an overnight SA run. We subsequently failed to replicate this with 200+ deliberate attempts from 4-11 seeds, orientation variants, and random starts. This suggests the 4-11 basin has a narrow entry funnel.

5.3 Oracle Cost Determines Method Choice

This study provides a clear example where algorithmic sophistication was counterproductive:

- Numba SA ($\sim$1M steps/sec): found $R = 2.976$.
- Greedy hill-climber ($\sim$120 eval/sec, exact): found $R = 2.974679$.
- MBH/PBH + L-BFGS-B ($\sim$100 iter/hr): found nothing.

The least sophisticated method with the most accurate oracle won. We conjecture the following heuristic for method selection based on oracle cost:

- Oracle cost < 1 ms: greedy hill-climbing with exact evaluation at every step.
- Oracle cost 1–100 ms: SA with periodic exact validation; approximate energy for candidate screening.
- Oracle cost > 100 ms: surrogate models, population methods, approximate energy throughout.

5.4 Leaderboard Context

As of March 28, 2026: our score $R = 2.97468$ ranked #9. A cluster at positions #5–#8 achieved $R \approx 2.9727$–2.9728 (gap: 0.002). The leader achieved $R = 2.96175$ (gap: 0.013). The 0.002 gap may be bridgeable by continuous refinement; the 0.013 gap likely requires a topological change.

6 Conclusions

We have presented the first computational study of packing 15 unit semicircles into a minimal enclosing circle, achieving $R \approx 2.975$. Our principal findings are:

1. **The 4-11 ring topology dominates.** Four inner semicircles at $r \approx 0.7$–1.0 and 11 outer at $r \approx 2.0$–2.6 outperform all other tested topologies by $\sim$0.48 in radius.
2. **Exact oracles beat approximate gradients.** When feasibility can be evaluated exactly in $\sim$0.5 ms, a greedy hill-climber using the exact oracle outperforms L-BFGS-B with approximate phi-function energy. The phi-function's Φ_{PP} component is unreliable for non-anti-parallel flat faces, distorting the optimisation landscape.
3. **The landscape has narrow basins.** The best basin was discovered by serendipity and could not be replicated from 200+ deliberate starts. This suggests that for semicircle packing, global methods must be paired with extremely long run times or problem-specific basin-hopping strategies.
4. **No conjugate pairs in the best solution.** Despite theoretical expectations, zero antiparallel flat-face contacts appear, suggesting that for $N = 15$ the optimal regime avoids pair formation.

Open questions. What topology do the top-ranked solutions use? What is the true optimum for $N = 15$ (likely in $[2.88, 2.96]$)? Can the phi-function be corrected for bounded semicircular segments? Does flat-face contact appear in the global optimum?

Reproducibility. Code is available at https://github.com/CamArmstr/shape-packing-challenge.

Acknowledgements. Problem posed by benedictbrady via the shape-packing-challenge repository. Computational work performed on a Windows 11 / WSL2 machine with a 16-core Intel CPU.

References

Bernardetta Addis, Marco Locatelli, and Fabio Schoen. Efficiently packing unequal disks in a circle. *Operations Research Letters*, 36(1):37–42, 2008. doi: 10.1016/j.orl.2007.03.001.

Peter Brass, William Moser, and János Pach. *Research Problems in Discrete Geometry*. Springer, New York, 2005. Section 1.5: semicircle packing listed as open.

Nikolai Chernov, Yuriy Stoyan, Tatiana Romanova, and Alexander Pankratov. Phi-functions for 2D objects formed by line segments and circular arcs. *Advances in Operations Research*, 2012: Article ID 346358, 2012. doi: 10.1155/2012/346358.

László Fejes Tóth. Packing and covering. In *Handbook of Convex Geometry*. North-Holland, 1971. Semicircle packing density question posed; still open.

Andrea Grosso, Marco Locatelli, and Fabio Schoen. Solving the problem of packing equal and unequal circles in a circular container. *Journal of Global Optimization*, 47(1):63–86, 2010. doi: 10.1007/s10898-009-9458-3.

Eckard Specht. Packomania: The encyclopedia of circle packings. http://packomania.com, 2024. Contains circles-in-circle data; no semicircle-in-circle results.